

DECODE LABS INTERNSHIP

TASK 1: DATA CLEANING

1.Introduction

Data cleaning is one of the most important phases of the data analytics lifecycle. Real-world datasets often contain missing values, duplicate entries, inconsistent formatting, and inaccurate records that can negatively affect analysis results. Before performing any analysis, it is essential to clean and prepare the dataset.

In this task, the E-Commerce Orders dataset was examined and cleaned using Microsoft Excel to improve data quality and ensure reliable results for further analysis.

2.Objective

The objectives of this task were:

- Identify and remove duplicate records.
- Handle missing and inconsistent values.
- Standardize data formats.
- Validate numerical and date fields.
- Prepare a clean dataset for analysis and dashboard development.

3.Dataset Description

The dataset contains E-Commerce transaction records including:

- Order ID
- Order Date
- Customer ID
- Product Name
- Quantity
- Unit Price
- Payment Method

- Referral Source
- Total Price

Total Records Analysed: 1200

AutoSave

Cleaned_Dataset_Ayushixls... • Last Modified: 2h ago

Search

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelpTable Design

Copypaste

Format Painter

Clipboard

Calibri

11

A

A⁺

Figure 1: Clean Dataset

4.Tools Used

- Microsoft Excel
- Sort & Filter
- Remove Duplicates
- Data Validation
- Conditional Formatting

5.Data Cleaning Process

- Duplicate Removal

The dataset was checked for duplicate rows using Excel’s Remove Duplicates feature. Duplicate records were removed to ensure each transaction appeared only once.

- Missing Value Treatment

Blank and incomplete fields were identified and corrected wherever possible. Records with insufficient information were reviewed carefully.

- **Data Type Validation**

Columns containing dates, quantities, and prices were checked to ensure correct formatting.

- **Standardization**

Column names and categorical values were standardized to maintain consistency throughout the dataset.

- **Revenue Verification**

Revenue calculations were verified to ensure Total Price values matched Quantity × Unit Price.

Change ID	Description	Impact	Status
CR001	Filled blank CouponCode with "No Coupon"	309 missing values fixed	Resolved
CR002	Checked duplicate OrderID	0 duplicates found	Resolved
CR003	Standardized Date format to yyyy-mm-dd	Dates cleaned	Resolved
CR004	Cleaned text columns using TRIM/PROPER	Text consistency improved	Resolved

Figure 2: Change Log Sheet

6.Results

After cleaning:

- Dataset consistency improved.
- Duplicate records were removed.
- Data accuracy increased.

- Dataset became suitable for analysis.

7.Conclusion

Data cleaning significantly improved the quality of the dataset and provided a reliable foundation for Exploratory Data Analysis, SQL querying, and dashboard development.